

LECTURE 8.2

Dynamic Heat Transfer

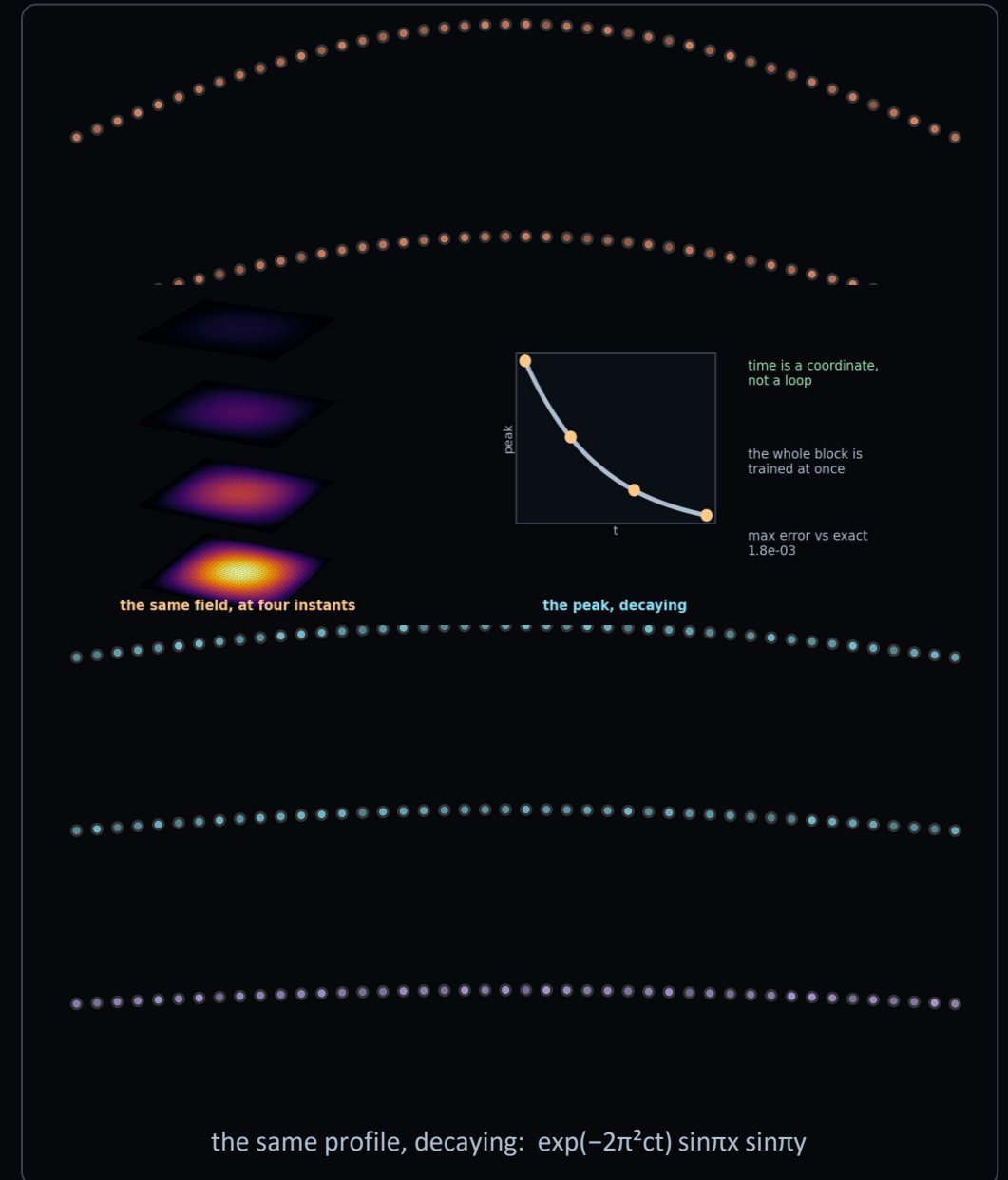
How fast does a device heat up, and what happens under a changing load? Same plate, same hole, same code — with time.

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Aalborg University · ret@et.aau.dk · Deep Learning for Engineering · 2026

Liu (2025), PINN with Python, Ch. 4



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **size** — a time window from the Fourier number, before you sample anything
- **build** — a trial solution that holds the initial condition and the boundary at once
- **explain** — why the loss does not know which way time runs
- **read** — a time history, and name the three failures visible in it
- **identify** — a diffusivity from a cooling curve, and say why steady state cannot
- **recognise** — stiffness, and say why a single time window cannot serve two scales

WHAT TODAY ASSUMES

from L8.1	the plate, the hole, the conservative form
from L7.2	the space-time domain, and hard IC
from L7.1	residual, composite loss, two-stage training

ONE TERM, ONE CONDITION

The equation gains a time derivative and the problem gains an initial condition. Everything else is L8.1, unchanged.

AND ONE NEW FAILURE

Causality. Nothing in the loss requires early times to be right before late ones.

One Term, One Condition, One New Failure Mode

	L8.1 — STATIONARY	L8.2 — DYNAMIC
The operator	$\nabla^2 T + Q/k = 0$	$\partial T / \partial t = \alpha \nabla^2 T + Q/\rho c_p$
Network inputs	x, y	x, y, t
Conditions	<i>boundary only</i>	<i>boundary + one initial</i>
Domain	Ω	$\Omega \times (0, t_{en}^d]$
New failure mode	—	<i>causality</i>
The question	<i>how hot?</i>	<i>how fast, and under what duty cycle?</i>

Nothing about the network, the optimiser or the sampling strategy changes. The whole lecture is about the consequences of those three cells in the right-hand column.

The Transient Conduction Equation

ENERGY BALANCE IN A CONTROL VOLUME

$$\rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + Q(x, y, t)$$

THERMAL DIFFUSIVITY

$$\alpha = \frac{k}{\rho c_p} \quad [m^2/s]$$

THE FORM WE SOLVE

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{Q}{\rho c_p}$$

- **the new term is stored energy** — heat that raises the temperature rather than moving on
- divide through and one material parameter remains: the diffusivity
- **its units are metres squared per second** — a length-to-time relationship
- **copper 1.1e-4, silicon 8.8e-5, alumina far lower** — a spread that matters

How Long Is a Transient?

FOURIER NUMBER

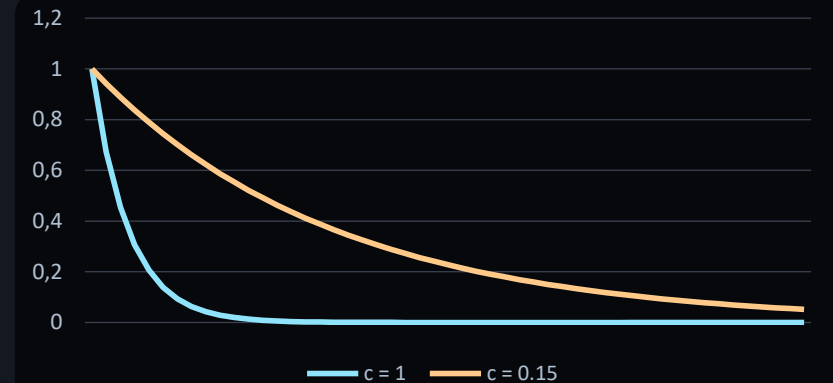
$$Fo = \frac{\alpha t}{L^2}$$

THERMAL TIME CONSTANT OF THE LOWEST MODE

$$\tau = \frac{L^2}{\pi^2 \alpha} \Rightarrow \text{steady state at } t \gtrsim 4\tau$$

- the Fourier number is dimensionless time
- far below one and the transient has barely started; near one and it is nearly over
- so it tells you where to put the end of your time window
- too small and you never reach steady state; too large and you waste points
- a silicon die reaches steady state in tens of milliseconds

AMPLITUDE DECAY



most of the physics happens in the first 10% of the window

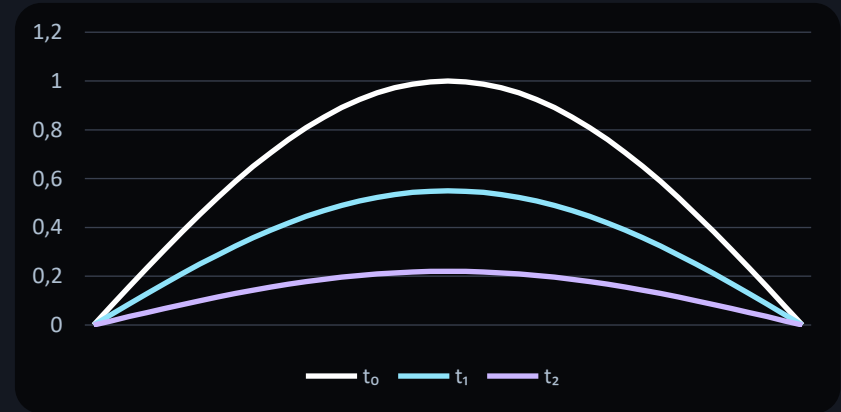
A Benchmark with a Closed-Form Answer

EXACT SOLUTION: ZERO SOURCE, ZERO DIRICHLET ON ALL EDGES, SINUSOIDAL IC

$$u_{exact}(x, y, t) = \exp(-2\pi^2 ct) \sin(\pi x) \sin(\pi y)$$

- **separation of variables** — the spatial factor is the lowest Helmholtz mode
- the shape never changes, only its amplitude
- which is what makes it a clean benchmark
- the same solution as Ex_7.2, so results compare directly across lectures
- **verify it before trusting it** — substitute and confirm the residual is zero

SHAPE FIXED, AMPLITUDE DECAYING



a single mode decays without changing shape

Time Is a Coordinate, Not a Loop

DOMAIN

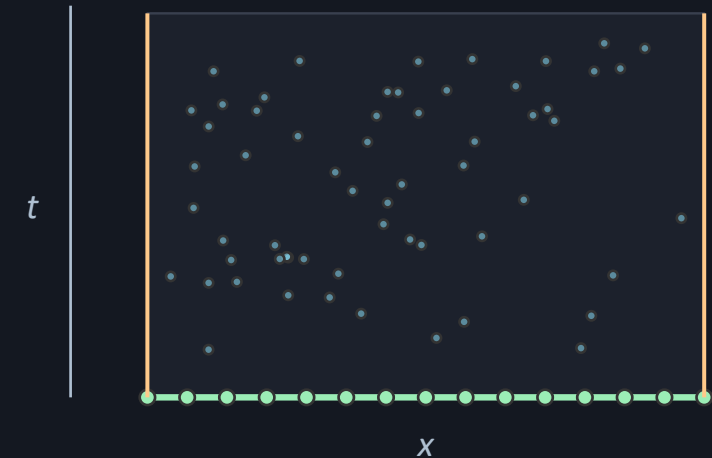
$$(x, y, t) \in \Omega \times (0, t_{end}] \subset \mathbb{R}^3$$

- a classical solver marches: each step computed from the last
- **a PINN does not** — time joins x and y as a third input
- the whole block is sampled at once
- the benefit is a solution continuous in t , differentiable with respect to it
- the cost is that nothing enforces the arrow of time

ONE INITIAL CONDITION

$$u(x, y, 0) = f_{IC}(x, y)$$

ONE BLOCK, SAMPLED AT ONCE



green: IC at $t = 0$ · amber: DBC for all t
 the IC occupies one face — a measure-zero slice

Hard-Enforcing the Initial Condition

TRIAL SOLUTION ENFORCING THE IC ALONE

$$u_{trial}(x, y, t) = (1 - t) f_{IC}(x, y) + t \mathcal{N}_{\hat{w}}(x, y, t)$$

- at $t = 0$ the second term vanishes and the initial condition is returned exactly
- so that loss term disappears
- the boundary conditions remain soft in this form
- **the $(1 - t)$ factor assumes t in $[0,1]$** — non-dimensionalise first
- and note what it does to causality: the solution is anchored structurally

FACTOR BY FACTOR

$$(1 - t) f^{IC}(x, y)$$

= f^{IC} at $t = 0$

fades out as t grows

$$t \cdot \mathcal{N}_{\hat{w}}(x, y, t)$$

= 0 at $t = 0$

free for $t > 0$

The two terms hand over: the initial state fades out exactly as the network fades in.

Enforcing the Initial and Boundary Conditions Together

BOTH CONDITIONS BUILT INTO THE ANSATZ

$$u_{trial}(x, y, t) = (1 - t) f_{IC}(x, y) + t \mathcal{N}_{\hat{w}}(x, y, t)$$

- **read the factors separately** — one holds $t = 0$, the other holds the boundary
- two loss terms disappear at once
- the composite loss collapses to the residual alone
- Liu implements this inside the PINN class, so the rest of the code never sees it
- **the two factors must agree at the corner** — check that first

WHAT REMAINS OF THE LOSS

$$\mathcal{L} = \mathcal{L}_{PDE}$$

CONSISTENCY CHECK

$f^{lc} = \sin(\pi x)\sin(\pi y)$ is zero on all four edges, so it agrees with the Dirichlet condition at $t = 0$. A constant initial temperature would not — and that mismatch is a common source of trouble.

Three Ways to Build the Loss

All soft

$$\mathcal{L}_p^d + \mathcal{L}^{\text{DBc}} + \mathcal{L}^{\text{IC}}$$

Three competing terms.
The IC is outnumbered by interior points and is the one usually lost.
Most flexible, least reliable.

IC enforced

$$\mathcal{L}_p^d + \mathcal{L}^{\text{DBc}}$$

Liu Ch. 4.5.
Causality risk removed;
boundary still penalised.
Use when the geometry is awkward but the IC is simple.

Both enforced

$$\mathcal{L}_p^d$$

Liu Ch. 4.4.
A single-term loss.
Most accurate and most stable — but needs a multiplier for the geometry.

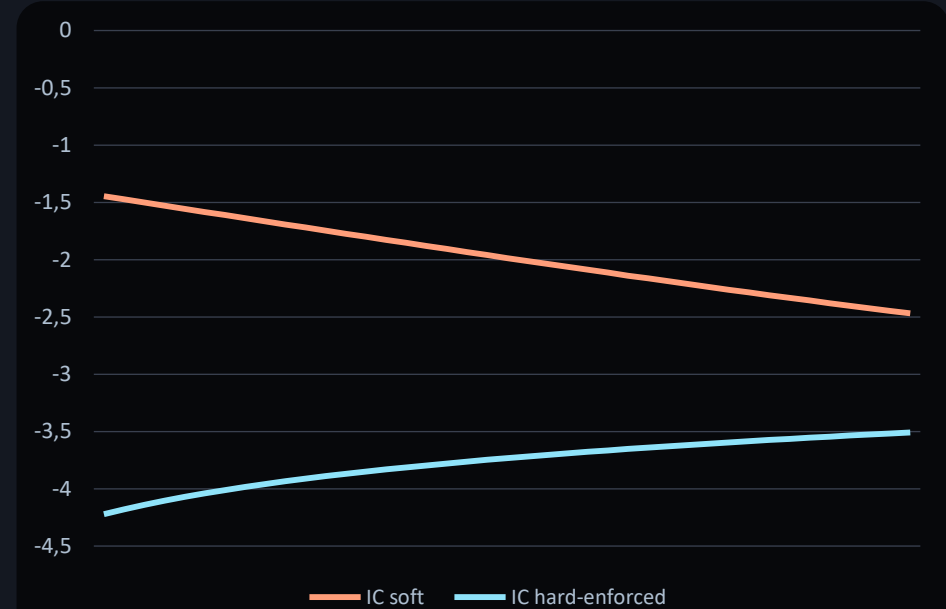
- three ways to build the loss, and their accuracy ordering is predictable
- all soft: everything competes
- IC hard: one competitor removed

- IC and BC hard: one term left, and nothing to balance

The Arrow of Time Is Not in the Loss

- every collocation point is treated identically, at $t = 0.01$ or $t = 0.99$
- so a PINN can satisfy the PDE at late times while the early transient is wrong
- the mechanism is a sampling imbalance, not a pathology
- the initial condition lives on a slice; the residual fills the block
- and it is worst exactly where transients matter most

ERROR AGAINST TIME



$\log_{10}$ relative error against t . With a soft IC the largest error sits at $t = 0$ — the one instant that was known exactly before training began.

Four Remedies, in Order of Preference

HARD-ENFORCE THE IC 1

The trial solution of slide 7 makes $t = 0$ exact by construction.
Removes the problem rather than mitigating it.

First choice. Costs nothing and needs no tuning.

CAUSAL WEIGHTING 2

Weight each time slab by the residual accumulated before it, so late times are only emphasised once earlier ones have converged.

Effective, but adds a hyperparameter ϵ .

TIME-DOMAIN DECOMPOSITION 3

Split $[0, t_{\text{en}}^d]$ into windows and train them in sequence, each starting from the previous window's end state.

Strong for long or stiff transients. More code.

DENSER EARLY SAMPLING 4

Concentrate collocation points where the solution changes fastest — usually the first 10% of the window.

Cheap and always sensible, but a partial fix.

$$w(t) = \exp\left(-\epsilon \int_0^t \mathcal{L}_{PDE}(s) ds\right)$$

Computing the Transient Residual

RESIDUAL

$$\mathcal{F}_{\hat{w}} = \frac{\partial T}{\partial t} - \alpha(T_{,xx} + T_{,yy}) - \frac{Q}{\rho c_p}$$

- three derivatives from one graph
- the time derivative is first order, so the graph is shallower than for the wave equation
- **order matters for efficiency** — take the first gradient once and slice it
- and evaluate the residual on the trial solution, not the raw network

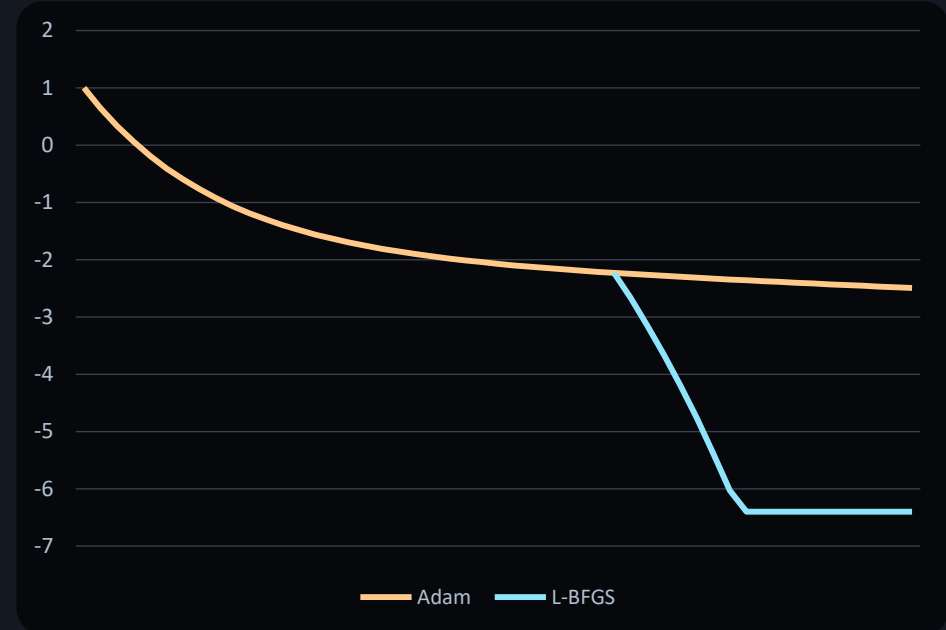
IN THE BOOK'S STYLE

```
T = trial(model, xyt)
g = grad(T, xyt)          # one pass
T_t = g[:, 2:3]
T_xx = d2(T, xyt, 0)
T_yy = d2(T, xyt, 1)
res = T_t - alpha*(T_xx + T_yy) - q
loss = (res**2).mean()
```

What the Benchmark Achieves

- with both conditions hard-enforced the benchmark reaches about $1e-06$
- **the loss history has the familiar shape** — Adam falls, then L-BFGS finishes
- read the error as a function of time, not as a single number
- a model can look excellent on the average and be wrong early
- plot contours at several instants and check the shape is preserved

LOSS HISTORY (LOG SCALE)

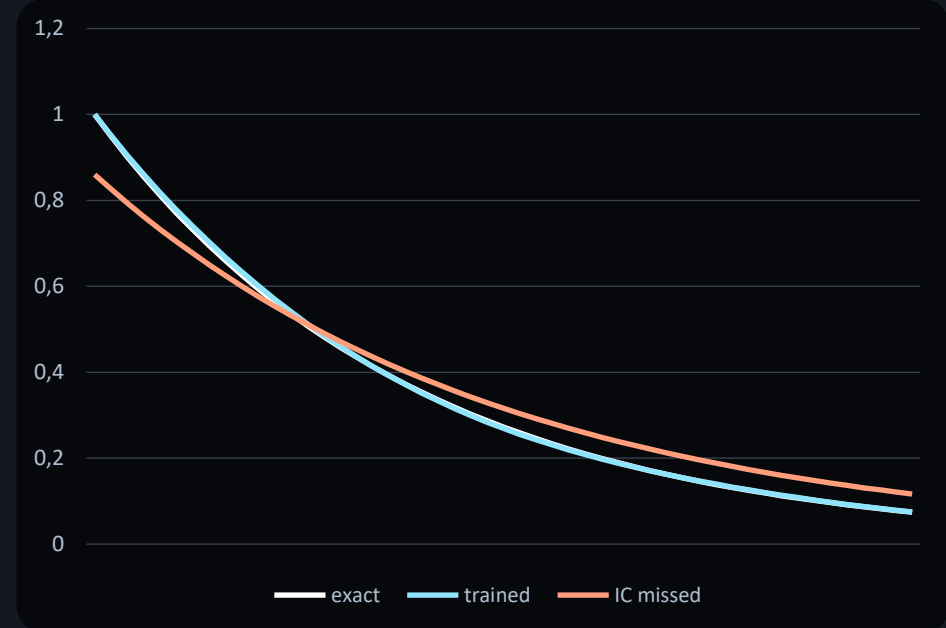


From about 10^1 down to 10^{-6} in roughly ten seconds. The handoff, not the Adam stage, is where the accuracy appears.

Reading the Time History

- the most useful diagnostic: pick a point, sweep t , plot the history
- three failure signatures are visible immediately
- an offset at $t = 0$ means the initial condition was not enforced
- the model is continuous in t , so the history costs one forward pass
- **and for a device this is the deliverable** — the junction temperature over time

CENTRE-POINT HISTORY



The orange curve starts below the exact value and decays too slowly — both symptoms of an initial condition that was penalised rather than enforced.

Does the Activation Choice Matter Here?

- L7.2 argued periodic activations suit oscillatory solutions
- a decaying transient is not oscillatory, so that argument does not apply
- **the requirement is unchanged** — a usable second derivative
- among the smooth candidates the differences here are modest
- match the inductive bias to the solution, not to the method

SOLUTION CHARACTER DECIDES

Decaying transient

smooth, monotone

tanh

Oscillatory (wave)

matches the basis

sin

Sharp front

sampling, not activation

tanh + adaptive

Any 2nd-order PDE

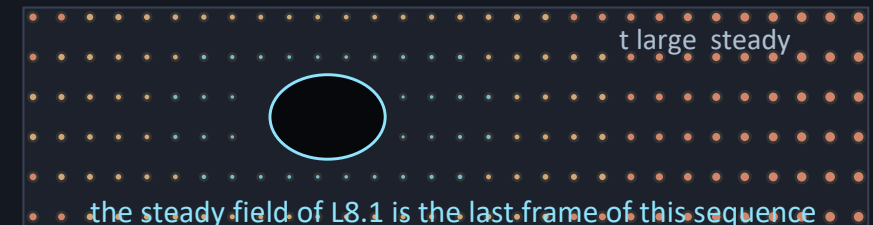
$\sigma'' = 0$ a.e.

not ReLU

The Same Plate, Now Transient

- identical geometry to L8.1 — the same plate, the same hole
- the only change is that the field now evolves
- and the engineering question changes with it: not how hot, but how long
- the sampling machinery carries over unchanged
- the hole condition can still be hard-enforced by the level set

THREE INSTANTS



Flux Conditions That Change With Time

A PRESCRIBED FLUX MAY DEPEND ON TIME AS WELL AS POSITION

$$-k \frac{\partial T}{\partial n} = \bar{q}(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega_N$$

- **nothing structural changes** — the normal derivative still comes from autograd
- but boundary points must now be sampled across time as well as along the boundary
- watch the point budget: boundary points times instants grows quickly
- **and this is how a duty cycle enters** — a switching load is a time-dependent source

A SWITCHING LOAD

$$Q(t) = Q_0 \Pi(t; t_{on}, t_{period})$$

DUTY CYCLE



Tuning the Weights, and When Not To

- Liu's default remains one for every term
- **diagnose before tuning** — plot the per-term losses separately
- prefer removing a term to weighting it
- every condition you hard-enforce is one you never have to balance
- scaling first, weights last

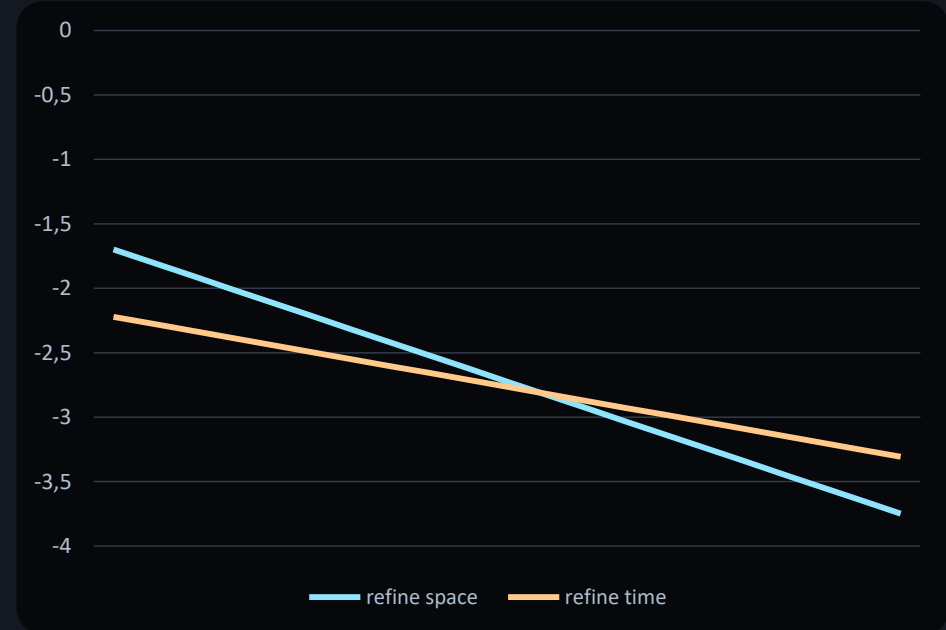
DECISION ORDER

- 1 Non-dimensionalise**
removes most imbalance at the source
- 2 Hard-enforce what you can**
a removed term needs no weight
- 3 Plot per-term losses**
diagnose before you act
- 4 Weight only what is ignored**
and report the value you used

Convergence in Space and in Time

- a transient has two refinement axes, and they behave differently
- the time direction is cheap for a smooth decay
- **the space direction follows the usual rule** — refine until the error plateaus
- and if the error is dominated by early times, refinement in space will not help
- that is a causality problem, not a resolution problem

REFINING SPACE VS. TIME



$\log_{10}$ relative L_2 against total point count. Both plateau — and the plateau, not the slope, is what tells you where the budget should go next.

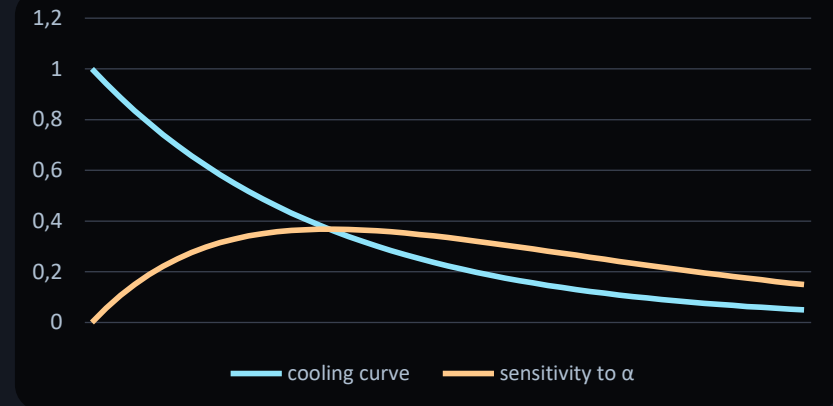
Identifying Diffusivity from a Transient

DIFFUSIVITY PROMOTED TO A TRAINABLE PARAMETER

$$\alpha \text{ trainable}, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

- a transient carries far more information about the material than a steady state
- in steady state the conductivity and the source are hard to separate
- the decay rate is set directly by the diffusivity
- so one thermocouple recording a cooling curve can recover it
- identify from the early transient, where the signal is

WHERE THE INFORMATION IS

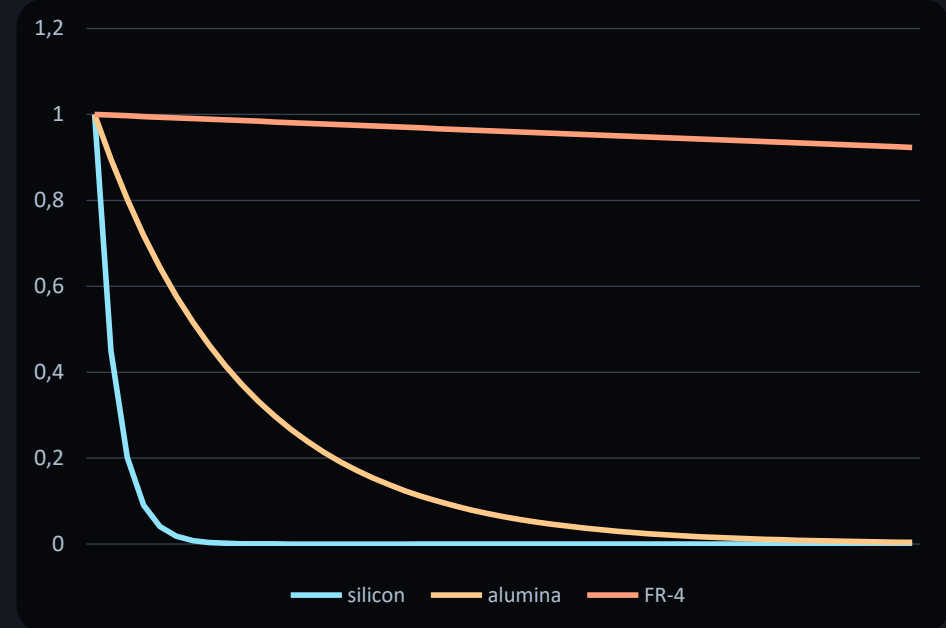


sensitivity peaks early, then vanishes

Stiffness: When Scales Separate

- a real assembly contains diffusivities differing by three orders of magnitude
- the die responds in milliseconds, the board in seconds
- a single time window cannot serve both
- sized for the board it under-resolves the die entirely
- this is stiffness, and it is a genuine limitation rather than a tuning problem

THREE MATERIALS, ONE PLOT



On a window that resolves silicon, FR-4 has not begun to respond. On one that captures FR-4, the silicon transient is a single collocation point.

Pitfalls Specific to Transient Heat

Causality violation

Late times fit before early ones, because the IC lives on a measure-zero slice.

Symptom: error largest at $t = 0$. Fix: hard-enforce the IC.

IC inconsistent with BC

A uniform initial temperature contradicts a zero Dirichlet edge at the corner $t = 0$.

Symptom: persistent error along the boundary at early times.

Time window mis-sized

t_{en}^d chosen without computing the thermal time constant first.

Symptom: either no steady state reached, or most points wasted on a flat field.

Late-time metric illusion

Relative error computed where the exact solution has decayed to $\sim 10^{-8}$.

Symptom: alarming relative errors that mean nothing. Fix: report absolute error late.

Stiff multi-material

Diffusivities separated by orders of magnitude in one domain.

Symptom: one material solved, the other untouched. Often needs a classical solver.

The fourth is worth stressing: at late times the benchmark amplitude is $\sim 5 \times 10^{-9}$, so a relative error of 100% may correspond to an absolute error of 10^{-8} .

Verifying a Transient Solution

CHECK $T = 0$ FIRST

Evaluate the model at the initial instant and compare with f^{lc} directly. If the IC is hard-enforced this must be zero to machine precision — a non-zero value means the trial solution is wrong.

CHECK THE DECAY RATE

Fit an exponential to the centre-point history and compare its rate with $2\pi^2c$. A wrong rate points at the residual or the diffusivity, not at the training.

CHECK THE SHAPE

For a single-mode benchmark the spatial profile must not change with time. Normalise each instant by its peak and overlay them — they should coincide.

CHECK ENERGY

With insulated boundaries the total stored energy must be conserved; with cooling it must fall at the rate the boundary flux implies. Integrate with the nodal weights.

Run all four before reporting anything. Each catches a different class of error, and the first two take seconds.

What a Thermal Engineer Actually Wants

- ▶ Not a field. A number, with a margin: the peak junction temperature under the worst duty cycle, and how much headroom remains against the rated limit.
- ▶ A trained PINN is continuous and differentiable in every input, including time and any parameter you chose to expose. That makes three things nearly free that are expensive classically.

TIME TO LIMIT

Solve $T(t) = T_{\max}$ directly on the trained model. No repeated forward solves, no interpolation between stored steps.

DUTY-CYCLE SWEEP

Expose the on-time as an input and train once. The whole design space is then one forward pass per query.

DESIGN SENSITIVITY

$\partial T_{\text{peak}} / \partial (\text{hole position})$ comes from the same autograd that computed the residual — gradients for free.

For a single transient solve, a classical implicit solver is faster and better validated. The PINN earns its place when you need the answer as a differentiable function of the design — which is what the last two columns describe.

Questions

TEN TO TEST WHETHER TODAY LANDED

- what does the diffusivity set, and what are its units telling you?
- how do you choose the end of the time window?
- what does a PINN do that a time-marching solver does not?
- why does the loss not know which way time runs?
- what does the $(1 - t)$ factor assume, and what breaks if it is false?
- **an offset at $t = 0$ in the time history** — what does it mean?

AND FOUR MORE

seven

why does a transient identify the diffusivity when steady state cannot?

eight

why is refinement in space no help if early times are wrong?

nine

why does periodic activation not help here?

ten

two materials, three orders of magnitude apart. What now?

THE ONE WITH NO SETTLED ANSWER

Should causality be enforced in the loss, and what would it cost?

BEFORE L9.1

Ex_08.2 notebooks 01 to 03. Bring a time history at the hottest point.

Ex_8.2 — The Plate, With Time

TASKS

- 1 Solve the transient benchmark with a soft IC and plot the error against time.
- 2 Re-solve with the IC hard-enforced by the $(1-t)$ trial solution; compare the error at $t = 0$.
- 3 Compute the thermal time constant and justify your choice of t_{en}^d from it.
- 4 Add the cooling hole from Ex_8.1 and confirm the late-time field matches the L8.1 steady solution.
- 5 Recover an unknown diffusivity from a synthetic cooling curve at five sensor points.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why is the error largest at $t = 0$ with a soft IC? sl. 10

Why does the steady state appear as the last frame? sl. 16

Why identify α from a transient rather than a steady state? sl. 20

Why can a relative error of 100% be harmless at late times? sl. 22

When does stiffness make you reach for a classical solver? sl. 21, 24

Reuses the geometry module from Ex_8.1 with a time coordinate added — nothing is rebuilt.

L8 in One Picture

	L8.1 STATIONARY	L8.2 DYNAMIC
Operator $\mathcal{F}$	$\nabla^2 T + Q/k$	$\partial T/\partial t - \alpha \nabla^2 T - Q/\rho c_p$
Material parameter	k only	$\alpha = k/\rho c_p$
Conditions	DBC, NBC, Robin	the same, plus one IC
Hard enforcement	spatial multiplier	spatial multiplier $\times (1-t)$
Characteristic failure	pure-Neumann degeneracy	causality violation
Verification	manufactured solution	plus decay rate and shape
The inverse win	k from boundary flux	α from a cooling curve

One column is not harder than the other. It has one more term, and everything in the last three rows follows from it.

From Heat to the Rest of the Course

L9	Fluid Flow Prediction	Add convection to diffusion and the equation becomes nonlinear. Burgers first, then Navier–Stokes.
L10	Electrochemical Devices	Species transport is the same parabolic operator you have just solved — now coupled to an elliptic potential field through reaction kinetics.
L11	Robotics Camera Navigation	The physics-informed idea without a PDE: the constraint becomes a dynamic model.
L12	Power Grid Stability	Poisson on a graph, with stability constraints of the kind seen in the Helmholtz loss.
L13	MiniProject Workshop	Your own problem, with the machinery you now have.

Heat is the reference problem for all of them: elliptic and parabolic, with conditions of every type, on real geometry.

Key Takeaways

- 1 Time is a coordinate**
The whole space–time block is fitted at once. That buys a solution continuous and differentiable in t — and costs you the arrow of time.
- 2 Causality is a sampling problem**
The initial condition occupies a measure-zero slice and is outvoted from the first epoch. Hard-enforce it and the problem disappears.
- 3 α , not k , governs the transient**
Diffusivity sets the timescale. Compute the thermal time constant before choosing t_{en}^d — it tells you where the physics lives.
- 4 Read the error against time**
A single L_2 number hides the failure that matters. At late times report absolute error, not relative.
- 5 The transient carries the information**
 α is invisible in a steady state and plainly visible in a cooling curve. This is where a PINN beats a classical solver outright.

Next — L9: Fluid Flow Prediction. Convection joins diffusion, and the equation stops being linear.